

Problem 3 (Bulgarian solitaire), Cell 1: cyclic partitions and cycles

Problem 3 team

Official setting.

A partition of a positive integer n is a weakly decreasing sequence $\lambda = (\lambda_1, \dots, \lambda_s)$ of positive integers with $\lambda_1 + \dots + \lambda_s = n$. The shift $\mathcal{B}(\lambda)$ is the partition whose parts are the positive numbers among $\lambda_1 - 1, \dots, \lambda_s - 1$ together with one extra part equal to s . Call λ cyclic if $\mathcal{B}^i(\lambda) = \lambda$ for some $i \geq 1$, and let $d_{\mathcal{B}}(\lambda) = \min\{i \geq 0 : \mathcal{B}^i(\lambda) \text{ is cyclic}\}$, $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \text{ is a partition of } n\}$. Write $T_k = k(k+1)/2$ and $\delta_k = (k, k-1, \dots, 2, 1)$. Every $n \geq 1$ satisfies $T_{k-1} < n \leq T_k$ for exactly one k , which we call the rank of n .

Section 1 restates these definitions in an equivalent form, proves that $d_{\mathcal{B}}$ is well defined and that the rank exists and is unique (Lemma 1.2, where we also write $n = T_{k-1} + r$, $1 \leq r \leq k$), and fixes further notation.

Official statement of the cell.

Let $n = T_k$. Prove that for every partition λ of n there is an i with $\mathcal{B}^i(\lambda) = \delta_k$, and that δ_k is the only cyclic partition of n . Then let n be arbitrary of rank k , say $n = T_{k-1} + r$ with $1 \leq r \leq k$: determine all cyclic partitions of n , and determine the number of distinct cycles of $\mathcal{B}$ on the partitions of n . Prove both.

Status. **Solved** (complete proof; no computation is used).

Answer and where it is proved.

- $n = T_k$: δ_k is the only cyclic partition (Corollary 2.6(a)), and $\mathcal{B}^i(\lambda) = \delta_k$ for every $i \geq d_{\mathcal{B}}(\lambda)$ (Corollary 2.7).
- $n = T_{k-1} + r$, $1 \leq r \leq k$: the cyclic partitions are exactly $(k-1+e_{k-1}, k-2+e_{k-2}, \dots, 1+e_1, e_0)$ with $e_i \in \{0, 1\}$ and $\sum_i e_i = r$ (zero parts omitted), so there are $\binom{k}{r}$ of them (Theorem 2.5, Brandt [2]).
- On them $\mathcal{B}$ acts as rotation of a necklace with r black and $k-r$ white beads, so the number of distinct cycles is

$$N(k, r) = \frac{1}{k} \sum_{d \mid \gcd(k, r)} \varphi(d) \binom{k/d}{r/d} \quad (\text{Theorem 3.1; OEIS A037306 [6]}),$$

and every cycle length divides k .

The characterisation follows the argument of Griggs–Ho [4, §2] (Etienne’s idea [3]), made fully explicit with a potential function. The count uses Burnside’s lemma, proved here for the cyclic group.

1 Setting and notation

Throughout, $T_m = m(m+1)/2$ for integers $m \geq 0$, and $[P] \in \{0, 1\}$ denotes the truth value of a statement P . A *partition* λ of $n \geq 1$ (written $\lambda \vdash n$) is a finite non-increasing sequence of positive integers $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_\ell$ with sum n ; $\ell = \ell(\lambda)$ is its number of parts, and we put $\lambda_i = 0$ for $i > \ell(\lambda)$. Partitions are identified with multisets of positive integers.

Definition 1.1 (Bulgarian solitaire). For $\lambda \vdash n$ let $\mathcal{B}(\lambda)$ be the partition whose parts are $\ell(\lambda)$ together with the numbers $\lambda_i - 1$ for those $i \leq \ell(\lambda)$ with $\lambda_i \geq 2$. Since $\ell(\lambda) + \sum_i (\lambda_i - 1) = n$, we have $\mathcal{B}(\lambda) \vdash n$. Put $\mathcal{B}^0 = \text{id}$ and $\mathcal{B}^{i+1} = \mathcal{B} \circ \mathcal{B}^i$. A partition μ is *cyclic* if $\mathcal{B}^P(\mu) = \mu$ for some $P \geq 1$. For $\lambda \vdash n$, $d_{\mathcal{B}}(\lambda)$ is the least $i \geq 0$ such that $\mathcal{B}^i(\lambda)$ is cyclic, and $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \vdash n\}$. A partition $\lambda \vdash n$ with $d_{\mathcal{B}}(\lambda) = D_{\mathcal{B}}(n)$ is called *extremal*.

$d_{\mathcal{B}}(\lambda)$ is well defined: n has finitely many partitions, so $\mathcal{B}^a(\lambda) = \mathcal{B}^{a+P}(\lambda)$ for some $a \geq 0$, $P \geq 1$, and then $\mathcal{B}^a(\lambda)$ is cyclic. If μ is cyclic with $\mathcal{B}^P(\mu) = \mu$, then $\mathcal{B}^P(\mathcal{B}(\mu)) = \mathcal{B}(\mu)$, so $\mathcal{B}(\mu)$ is cyclic. Hence $\mathcal{B}^i(\lambda)$ is cyclic for every $i \geq d_{\mathcal{B}}(\lambda)$ and for no $i < d_{\mathcal{B}}(\lambda)$.

Lemma 1.2 (rank). *Every $n \geq 1$ can be written in exactly one way as $n = T_{k-1} + r$ with integers $k \geq 1$ and $1 \leq r \leq k$. We call k the rank of n ; n is triangular iff $r = k$ (then $n = T_k$).*

Proof. $T_0 = 0 < T_1 < T_2 < \dots$ is strictly increasing and unbounded, so there is exactly one $k \geq 1$ with $T_{k-1} < n \leq T_k$; then $r = n - T_{k-1}$ satisfies $1 \leq r \leq T_k - T_{k-1} = k$. Conversely $n = T_{k-1} + r$ with $1 \leq r \leq k$ forces $T_{k-1} < n \leq T_k$, i.e. this k . If $n = T_m$ ($m \geq 1$), then $T_{m-1} < n \leq T_m$, so $k = m$ and $r = T_m - T_{m-1} = m = k$; conversely $r = k$ gives $n = T_k$. $\square$

2 Diagrams, diagonals, rotation, and Brandt's theorem

This section follows Griggs–Ho [4, §2] (the idea goes back to Etienne [3]); we make their “left shifting decreases the diagonal index” precise with an explicit potential Φ .

Let $\mathbb{P} = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$; the *cell* $(i, j) \in \mathbb{P}$ has *row* i and *column* j . The *diagram* of λ is

$$Y(\lambda) = \{(i, j) \in \mathbb{P} : 1 \leq j \leq \ell(\lambda), 1 \leq i \leq \lambda_j\},$$

so column j has λ_j cells (columns are parts, as in [4]). $Y(\lambda)$ determines λ , since $\lambda_j = \#\{i : (i, j) \in Y(\lambda)\}$. Row 1 of $Y(\lambda)$ is $\{(1, j) : j \leq \ell(\lambda)\}$ and column 1 has λ_1 cells; in particular

$$(1, w+1) \in Y(\lambda) \implies (1, w) \in Y(\lambda) \quad (w \geq 1). \quad (1)$$

For a finite sequence $a = (a_1, \dots, a_m)$ of non-negative integers let $X_a = \{(i, j) : 1 \leq j \leq m, 1 \leq i \leq a_j\}$. If a is non-increasing, then $X_a = Y(\nu)$ where ν is the partition formed by the non-zero entries of a .

For $w \geq 1$ the w -th *diagonal* is $D_w = \{(i, j) \in \mathbb{P} : i + j = w + 1\}$; it has the w cells $d_w(j) = (w + 1 - j, j)$, $1 \leq j \leq w$ ($d_w(j)$ is the cell of D_w in column j). $\mathbb{P}$ is the disjoint union of the D_w . Put $\Delta_0 = \emptyset$ and $\Delta_m = D_1 \cup \dots \cup D_m$ for $m \geq 1$. Column j of Δ_m consists of the cells (i, j) with $i \leq m + 1 - j$, so $\Delta_m = Y(\delta_m)$ with $\delta_m = (m, m - 1, \dots, 1)$, and $|\Delta_m| = T_m$.

Define $\rho : \mathbb{P} \rightarrow \mathbb{P}$ by $\rho(i, j) = (i - 1, j + 1)$ if $i \geq 2$ and $\rho(1, j) = (j, 1)$.

Lemma 2.1. (a) $\rho(d_w(j)) = d_w(j + 1)$ for $1 \leq j < w$ and $\rho(d_w(w)) = d_w(1)$. Hence ρ maps each D_w bijectively onto itself, ρ is a bijection of $\mathbb{P}$, and for $m \geq 0$, $\rho^m(d_w(j)) = d_w(j')$ where $1 \leq j' \leq w$ and $j' \equiv j + m \pmod{w}$. In particular ρ^w is the identity on D_w . (b) ρ preserves the quantity $i + j$ of a cell.

Proof. (a) For $j < w$ the cell $d_w(j) = (w+1-j, j)$ has row $w+1-j \geq 2$, so $\rho(d_w(j)) = (w-j, j+1) = d_w(j+1)$; and $\rho(d_w(w)) = \rho(1, w) = (w, 1) = d_w(1)$. So ρ acts on D_w as the cyclic shift $j \mapsto j+1$ of column indices modulo w ; the rest follows. (b) $(i-1) + (j+1) = i+j$ and $j+1 = 1+j$. $\square$

Lemma 2.2 (one move). *Let $\lambda \vdash n$, $s = \ell(\lambda)$ and $a = (s, \lambda_1 - 1, \lambda_2 - 1, \dots, \lambda_s - 1)$. Then $\rho(Y(\lambda)) = X_a$, and $Y(\mathcal{B}(\lambda)) = X_{a^*}$, where a^* is the non-increasing rearrangement of a . If $s \geq \lambda_1 - 1$, then a is non-increasing and $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$.*

Proof. The cells of row 1 of $Y(\lambda)$ are $(1, j)$, $1 \leq j \leq s$; ρ sends them to $(j, 1)$, i.e. exactly onto the cells $(1, 1), \dots, (s, 1)$ of column 1. A cell $(i, j) \in Y(\lambda)$ with $i \geq 2$ satisfies $2 \leq i \leq \lambda_j$ and is sent to $(i-1, j+1)$; for fixed j these images are exactly the cells $(1, j+1), \dots, (\lambda_j - 1, j+1)$. Hence $\rho(Y(\lambda)) = X_a$. The parts of $\mathcal{B}(\lambda)$ are the non-zero entries of a , so $Y(\mathcal{B}(\lambda)) = X_{a^*}$. The entries of a after the first are non-increasing because λ is; so a is non-increasing iff $s \geq \lambda_1 - 1$, and then $a^* = a$. $\square$

For a finite set $X \subseteq \mathbb{P}$ put $\Phi(X) = \sum_{(i,j) \in X} (i+j)$.

Lemma 2.3. *For every finite sequence a of non-negative integers, $\Phi(X_{a^*}) \leq \Phi(X_a)$, with equality iff a is non-increasing.*

Proof. $\Phi(X_a) = \sum_j \sum_{i=1}^{a_j} (i+j) = \sum_j \binom{a_j+1}{2} + F(a)$ with $F(a) = \sum_j j a_j$. The first sum does not change when a is permuted. If $a_j < a_{j+1}$, exchanging these two entries changes F by $ja_{j+1} + (j+1)a_j - ja_j - (j+1)a_{j+1} = a_j - a_{j+1} < 0$. Starting from a , repeatedly exchange some adjacent pair with $a_j < a_{j+1}$. Each exchange lowers the non-negative integer F , so the process stops, and it stops at a sequence with no adjacent ascent, i.e. at the non-increasing rearrangement a^* (which is unique). If a is not non-increasing at least one exchange happens, so $F(a^*) < F(a)$; if a is non-increasing then $a = a^*$. $\square$

Corollary 2.4. *For $\lambda \vdash n$: $\Phi(Y(\mathcal{B}(\lambda))) \leq \Phi(Y(\lambda))$, with equality iff $\ell(\lambda) \geq \lambda_1 - 1$; in the case of equality $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$.*

Proof. By Lemma 2.1, ρ is injective and preserves $i+j$, so $\Phi(\rho(X)) = \Phi(X)$. With a as in Lemma 2.2: $\Phi(Y(\lambda)) = \Phi(X_a) \geq \Phi(X_{a^*}) = \Phi(Y(\mathcal{B}(\lambda)))$, with equality iff a is non-increasing (Lemma 2.3), i.e. iff $\ell(\lambda) \geq \lambda_1 - 1$, and then Lemma 2.2 gives $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$. $\square$

For $k \geq 1$ and $E \subseteq D_k$ let λ_E be the partition whose j -th part, $1 \leq j \leq k$, is $k-j + [d_k(j) \in E]$ (zero parts omitted). The sequence $(k-j + \varepsilon_j)_{j=1}^k$ with $\varepsilon_j \in \{0, 1\}$ is non-increasing (consecutive differences are $1 + \varepsilon_j - \varepsilon_{j+1} \geq 0$), so λ_E is a partition, $|\lambda_E| = T_{k-1} + |E|$, and $Y(\lambda_E) = \Delta_{k-1} \cup E$ (column j consists of the $k-j$ cells of Δ_{k-1} and, if $d_k(j) \in E$, the cell $d_k(j) = (k+1-j, j)$). Writing $e_{k-j} = [d_k(j) \in E]$ we get $\lambda_E = (k-1 + e_{k-1}, k-2 + e_{k-2}, \dots, 1 + e_1, e_0)$.

Theorem 2.5 (Brandt [2]; see also [1, 3, 4]). *Let $n = T_{k-1} + r$ with $1 \leq r \leq k$. A partition of n is cyclic if and only if it equals λ_E for some $E \subseteq D_k$ with $|E| = r$; equivalently, iff it has the form $(k-1 + e_{k-1}, \dots, 1 + e_1, e_0)$ with $e_i \in \{0, 1\}$ and $\sum_i e_i = r$ (zero parts omitted). Moreover $Y(\mathcal{B}(\lambda_E)) = \rho(Y(\lambda_E))$, i.e. $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$, and $\mathcal{B}^k(\lambda_E) = \lambda_E$.*

Proof. ($\Leftarrow$ and “moreover”). Let $\mu = \lambda_E$ with $E \subseteq D_k$, $|E| = r$. Row 1 of $Y(\mu) = \Delta_{k-1} \cup E$ consists of the cells $(1, w) = d_w(w)$ with $w \leq k-1$ and possibly $d_k(k)$; so $\ell(\mu) = k-1 + [d_k(k) \in E] \geq k-1$. Column 1 consists of $d_w(1)$ for $w \leq k-1$ and possibly $d_k(1)$; so $\mu_1 = k-1 + [d_k(1) \in E] \leq k$. Thus $\ell(\mu) \geq \mu_1 - 1$ and Corollary 2.4 and Lemma 2.1 give $Y(\mathcal{B}(\mu)) = \rho(\Delta_{k-1} \cup \rho(E)) = \Delta_{k-1} \cup \rho(E)$ with $\rho(E) \subseteq D_k$, $|\rho(E)| = r$. Hence $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$, and by induction $\mathcal{B}^k(\lambda_E) = \lambda_{\rho^k(E)} = \lambda_E$ because ρ^k is the identity on D_k . So μ is cyclic.

($\Rightarrow$). Let $\lambda \vdash n$ with $\mathcal{B}^P(\lambda) = \lambda$, $P \geq 1$, and write $\lambda^{(m)} = \mathcal{B}^m(\lambda)$. By Corollary 2.4, $\Phi(Y(\lambda^{(0)})) \geq \Phi(Y(\lambda^{(1)})) \geq \dots \geq \Phi(Y(\lambda^{(P)})) = \Phi(Y(\lambda^{(0)}))$, so all these are equalities and $Y(\lambda^{(m+1)}) = \rho(Y(\lambda^{(m)}))$ for $0 \leq m < P$. As $\lambda^{(m+P)} = \lambda^{(m)}$, this holds for all $m \geq 0$, and therefore $Y(\lambda^{(m)}) = \rho^m(Y(\lambda))$ for all $m \geq 0$.

Claim: there is no $w \geq 1$ such that $D_w \not\subseteq Y(\lambda)$ and $D_{w+1} \cap Y(\lambda) \neq \emptyset$. Suppose $d_w(\alpha) \notin Y(\lambda)$ and $d_{w+1}(\beta) \in Y(\lambda)$. Since $\gcd(w, w+1) = 1$, the Chinese remainder theorem gives $m \geq 0$ with $m \equiv w - \alpha \pmod{w}$ and $m \equiv w + 1 - \beta \pmod{w+1}$. By Lemma 2.1, $\rho^m(d_w(\alpha)) = d_w(w) = (1, w)$ and $\rho^m(d_{w+1}(\beta)) = d_{w+1}(w+1) = (1, w+1)$. As ρ^m is injective, $(1, w) \notin \rho^m(Y(\lambda)) = Y(\lambda^{(m)})$ while $(1, w+1) \in Y(\lambda^{(m)})$, contradicting (1).

$Y(\lambda)$ is finite, so some D_w is not contained in it; let $K \geq 1$ be the least such w . Then $\Delta_{K-1} \subseteq Y(\lambda)$. By the claim, $D_{K+1} \cap Y(\lambda) = \emptyset$; then $D_{K+1} \not\subseteq Y(\lambda)$, so again by the claim $D_{K+2} \cap Y(\lambda) = \emptyset$, and inductively $D_w \cap Y(\lambda) = \emptyset$ for all $w > K$. Thus $Y(\lambda) = \Delta_{K-1} \cup E'$ with $E' = Y(\lambda) \cap D_K \neq D_K$; let $r' = |E'| \in \{0, \dots, K-1\}$, so $n = T_{K-1} + r'$ and $Y(\lambda) = Y(\lambda_{E'})$ (with k replaced by K in the definition of $\lambda_{E'}$). If $r' \geq 1$, then $T_{K-1} < n < T_K$, so by Lemma 1.2 $K = k$, $r' = r$, and $\lambda = \lambda_{E'}$. If $r' = 0$, then $K \geq 2$ (as $n \geq 1$) and $n = T_{K-1} = T_{K-2} + (K-1)$, so $k = K-1$, $r = k$, and $Y(\lambda) = \Delta_k = \Delta_{k-1} \cup D_k$, i.e. $\lambda = \lambda_{D_k}$ with $|D_k| = k = r$. $\square$

Corollary 2.6. *Let $n = T_{k-1} + r$, $1 \leq r \leq k$.*

- (a) *If $r = k$ (i.e. $n = T_k$), the only cyclic partition of n is $\delta_k = (k, k-1, \dots, 1)$, and $\mathcal{B}(\delta_k) = \delta_k$.*
- (b) *Every cyclic $\mu \vdash n$ satisfies $Y(\mu) \subseteq \Delta_k$, $\Delta_{k-1} \subseteq Y(\mu)$, $\ell(\mu) \in \{k-1, k\}$ and $\mathcal{B}^k(\mu) = \mu$.*
- (c) *If $\mu \vdash n$ is cyclic, then for every $a \geq 0$: $\sum_{i=a}^{a+k-1} \ell(\mathcal{B}^i(\mu)) = k(k-1) + r$. If moreover $r < k$, both values $k-1$ and k occur among $\ell(\mathcal{B}^a(\mu)), \dots, \ell(\mathcal{B}^{a+k-1}(\mu))$.*

Proof. (a) and (b) are immediate from Theorem 2.5 ($E = D_k$ when $r = k$; $\ell(\lambda_E) = k-1 + [d_k(k) \in E]$ was computed in its proof). (c) Write $\mu = \lambda_E$. Then $\mathcal{B}^i(\mu) = \lambda_{\rho^i(E)}$, so $\ell(\mathcal{B}^i(\mu)) = k-1 + [d_k(k) \in \rho^i(E)] = k-1 + [d_k(j_i) \in E]$, where $d_k(j_i) = \rho^{-i}(d_k(k))$, i.e. $j_i \equiv k-i \pmod{k}$ (Lemma 2.1). As i runs through $a, \dots, a+k-1$, j_i runs through all of $1, \dots, k$ once, so the sum is $k(k-1) + |E| = k(k-1) + r$. If $0 < r < k$, some j_i lies in E and some does not. $\square$

Corollary 2.7. *Let $n = T_k$. For every partition λ of n , $\mathcal{B}^i(\lambda) = \delta_k$ for every $i \geq d_{\mathcal{B}}(\lambda)$; in particular there is an i with $\mathcal{B}^i(\lambda) = \delta_k$.*

Proof. $d_{\mathcal{B}}(\lambda)$ is finite (Section 1), and $\mathcal{B}^i(\lambda)$ is cyclic for every $i \geq d_{\mathcal{B}}(\lambda)$. By Corollary 2.6(a) the only cyclic partition of T_k is δ_k . $\square$

3 The cycles of $\mathcal{B}$ and necklaces

A *cycle* of $\mathcal{B}$ on the partitions of n is a set $\{\mathcal{B}^i(\mu) : i \geq 0\}$ with μ cyclic. For $E \subseteq D_k$ let $w(E) = (w_1, \dots, w_k) \in \{0, 1\}^k$ with $w_j = [d_k(j) \in E]$, and let $W_{k,r} = \{w \in \{0, 1\}^k : w_1 + \dots + w_k = r\}$. Let $\sigma(w_1, \dots, w_k) = (w_k, w_1, \dots, w_{k-1})$ (rotation).

Theorem 3.1. *Let $n = T_{k-1} + r$, $1 \leq r \leq k$, and let $\mathcal{C}_n$ be the set of cyclic partitions of n .*

- (a) *$\lambda_E \mapsto w(E)$ is a well-defined bijection $\mathcal{C}_n \rightarrow W_{k,r}$; in particular $|\mathcal{C}_n| = \binom{k}{r}$.*
- (b) *If $\mu = \lambda_E$, then $\mathcal{B}(\mu) = \lambda_{\rho(E)}$ and $w(\rho(E)) = \sigma(w(E))$. Hence $\mathcal{B}$ permutes $\mathcal{C}_n$.*
- (c) *The cycles of $\mathcal{B}$ are exactly the images of the σ -orbits on $W_{k,r}$ (necklaces with r black and $k-r$ white beads). Their number is $N(k, r) = \frac{1}{k} \sum_{d \mid \gcd(k, r)} \varphi(d) \binom{k/d}{r/d}$.*

- (d) The cycle through λ_E has length equal to the least $p \geq 1$ with $\sigma^p(w(E)) = w(E)$, and this length divides k .

Proof. (a) By Theorem 2.5, $E \mapsto \lambda_E$ maps the r -subsets of D_k onto $\mathcal{C}_n$. It is injective, because λ_E determines $Y(\lambda_E) = \Delta_{k-1} \cup E$, and $E = Y(\lambda_E) \cap D_k$. Finally $E \mapsto w(E)$ is a bijection from r -subsets of D_k onto $W_{k,r}$.

(b) By Theorem 2.5, $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$. By Lemma 2.1, $\rho^{-1}(d_k(j)) = d_k(j-1)$ for $j \geq 2$ and $\rho^{-1}(d_k(1)) = d_k(k)$. So $w(\rho(E))_j = [\rho^{-1}(d_k(j)) \in E] = w(E)_{j-1}$ (with $w_0 := w_k$), i.e. $w(\rho(E)) = \sigma(w(E))$. Since σ is a bijection of $W_{k,r}$, $\mathcal{B}$ is a bijection of $\mathcal{C}_n$.

(c) By (b), for $\mu = \lambda_E$ the cycle $\{\mathcal{B}^i(\mu) : i \geq 0\}$ corresponds to the forward orbit $\{\sigma^i(w(E)) : i \geq 0\}$, which is the orbit under the group generated by σ because $\sigma^k = \text{id}$. To count orbits, let the group $G = \mathbb{Z}/k\mathbb{Z}$ act on $W_{k,r}$ by $m \cdot w = \sigma^m(w)$.

Burnside's lemma for G : the number of orbits is $\frac{1}{k} \sum_{m=0}^{k-1} |\text{Fix}(\sigma^m)|$. Indeed, for $w \in W_{k,r}$ the stabiliser $S_w = \{m \in G : \sigma^m w = w\}$ is a subgroup. The map $G \rightarrow \text{Orb}(w)$, $m \mapsto \sigma^m(w)$, is onto, and its fibres are the cosets of S_w , so $|\text{Orb}(w)| |S_w| = k$. Hence $\sum_m |\text{Fix}(\sigma^m)| = \sum_w |S_w| = \sum_O \sum_{w \in O} k/|O| = k \cdot \#\{\text{orbits}\}$.

Fixed points: let $0 \leq m \leq k-1$ and $g = \gcd(m, k)$ (so $g = k$ for $m = 0$). $\sigma^m w = w$ means $w_j = w_{j-m}$ for all j (indices mod k), i.e. w is constant on the cosets of the subgroup generated by m in $\mathbb{Z}/k\mathbb{Z}$. This subgroup equals $g\mathbb{Z}/k\mathbb{Z}$: g is a multiple of m modulo k by Bézout, and m is a multiple of g . So w is constant on the g residue classes mod g , each of size k/g , and w is determined by $(w_1, \dots, w_g)$, with $|w| = (k/g)(w_1 + \dots + w_g)$. With $d = k/g$ we get $|\text{Fix}(\sigma^m)| = \binom{k/d}{r/d}$ if $d \mid r$, and 0 otherwise.

Counting m : for $d \mid k$, the $m \in \{0, \dots, k-1\}$ with $\gcd(m, k) = k/d$ are the $m = (k/d)m'$ with $0 \leq m' < d$ and $\gcd(m', d) = 1$. There are $\varphi(d)$ of them ($m' = 0$ counts only when $d = 1$, and $\varphi(1) = 1$). Therefore

$$\#\{\text{orbits}\} = \frac{1}{k} \sum_{d \mid k, d \mid r} \varphi(d) \binom{k/d}{r/d} = N(k, r).$$

(d) The length of the cycle is $|\text{Orb}(w(E))| = k/|S_{w(E)}|$. The set $\{m \in \mathbb{Z} : \sigma^m(w(E)) = w(E)\}$ is a subgroup of $\mathbb{Z}$ containing k , hence equals $p\mathbb{Z}$ for the least $p \geq 1$ with $\sigma^p(w(E)) = w(E)$, and $p \mid k$. Its image $S_{w(E)}$ in $\mathbb{Z}/k\mathbb{Z}$ has k/p elements, so the cycle length is p , a divisor of k . $\square$

Corollary 3.2. For $r \in \{1, k-1, k\}$ there is exactly one cycle: $N(k, 1) = N(k, k-1) = \binom{k}{1}/k = 1$ (as $\gcd(k, 1) = \gcd(k, k-1) = 1$), and $N(k, k) = 1$ because $|W_{k,k}| = 1$. For $r = k$ it is the fixed point δ_k . For $r \in \{1, k-1\}$ and $k \geq 2$ it has length k : a word with a single 1 (or a single 0) is moved by σ^m for $0 < m < k$, so by Theorem 3.1(d) its period is k .

4 Computational check

The script `bs.py` enumerates all partitions of every $n \leq 60$ and builds the functional graph of $\mathcal{B}$ from the definition. It finds the cycles of that graph independently of the theorems above and checks that: (i) the set of cyclic partitions equals Brandt's list; (ii) the number of cycles equals $N(k, r)$; (iii) every cycle length divides k . All checks pass (python3 `bs.py` 60, about 1.5 minutes). These computations are a sanity check only; the proofs above do not use them.

References

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